

Implement a simulation of a modified 2048 game played on a 16×16 grid.

The program receives a sequence of moves and must simulate the board state according to standard 2048 rules.

After every move, the current board state must be printed.

Game Rules

1. Board

Grid size: 16×16

Each cell contains:

0 $\rightarrow$ empty

Power of 2 $\rightarrow$ tile value

2. Allowed Moves

UP

DOWN

LEFT

RIGHT

Each move shifts tiles completely toward that direction.

3. Movement Behavior

When a move occurs:

Tiles slide as far as possible.

Equal adjacent tiles merge.

Each tile merges only once per move.

After merging, tiles shift again to remove gaps.

4. Merge Examples

$[2, 2, 0, 0] \rightarrow \text{LEFT} \rightarrow [4, 0, 0, 0]$

$[2, 2, 2, 0] \rightarrow \text{LEFT} \rightarrow [4, 2, 0, 0]$

$[4, 4, 4, 4] \rightarrow \text{LEFT} \rightarrow [8, 8, 0, 0]$